Uveal Melanoma Radiation

^{125}I Brachytherapy Versus Helium Ion Irradiation

DEVRON H. CHAR, MD,[1,2] JOSEPH R. CASTRO, MD,[3] JEANNE M. QUIVEY, MD,[2] THEODORE L. PHILLIPS, MD,[2,3] ALEXANDER R. IRVINE, MD,[1] ROBERT D. STONE, MD,[1] STEWART KROLL, MA[1]

Abstract: The optimum radiation therapy for uveal melanoma is uncertain. Both helium ion irradiation and ^{125}I brachytherapy have been used to treat this neoplasm. This investigation analyzed the control and complication rates of uveal melanomas treated with helium ions or ^{125}I plaques. In both a retrospective and a prospective dynamically balanced study, the control rates appeared to be similar. There were more posterior segment complications after ^{125}I plaques and more anterior segment complications, including neovascular glaucoma, after helium ion irradiation. The follow-up period is too short to draw definitive conclusions on the radiation complications. Overall, approximately 89% of eyes were retained and less than 4% of treated eyes were removed because of failure to control the tumor. *Ophthalmology 96:1708–1715, 1989*

Alternative radiations to treat uveal melanoma and preserve the eye have been attempted since 1929.[1,2] Most early radiation brachytherapy was done with ^{60}Co radioactive plaques. Mainly small tumors were treated with good local control; however, a number of radiation complications were noted. Newer brachytherapy and charged particle radiation techniques were implemented to attempt to treat larger tumors and decrease ocular morbidity.[3-13] Over 4000 eyes with uveal melanomas have been treated with these radiation techniques; however, a number of issues regarding the relative values of different radiation delivery systems remain unresolved.

Relative advantages and disadvantages of charged particles (helium ions or protons) versus radioactive plaques for uveal melanoma therapy are poorly delineated.[3] Brachytherapy and treatment with charged particles differ in focusing ability, dose homogeneity, dose rate, biologic efficacy, ability to shield contiguous structures from radiation, availability, and cost. Although there are theoretic advantages associated with each modality, the relative importance of some theoretic factors may be lessened by at least two factors.[13] First, approximately 50% of uveal melanomas occur less than 3 mm from the optic nerve or fovea; in order to ensure adequate treatment margins, most of these eyes receive maximal radiation dose to both the tumor and these visually vital structures regardless of the radiation delivery system. Second, soluble, intrinsic tumor, angiogenic factors, especially in large melanomas, and lymphokines from inflammatory cells that infiltrate regressing tumors probably are as important as radiation effect in causing ocular morbidity.[14–16]

We studied the relative morbidities and efficacies of helium ion and ^{125}I plaque irradiations in both a retrospective analysis of a phase I and II study and an ongoing, dynamically balanced, prospective randomized trial comparing ^{125}I brachytherapy and helium ion irradiations.

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[1] Ocular Oncology Unit, Department of Ophthalmology, University of California, San Francisco.
[2] Department of Radiation Oncology, University of California, San Francisco.
[3] Radiotherapy Section, Lawrence Berkeley Laboratory, University of California, Berkeley.

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Reprint requests to Devron H. Char, MD, Ocular Oncology Unit, PO Box 0730, University of California, San Francisco, CA 94143.